

Hall Ticket No. -

Code: R123926

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) II Semester (Main) Examinations-September-2023

Engineering Mechanics

Time: 03 Hours

Max. Marks-60

- Note: i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

PART A

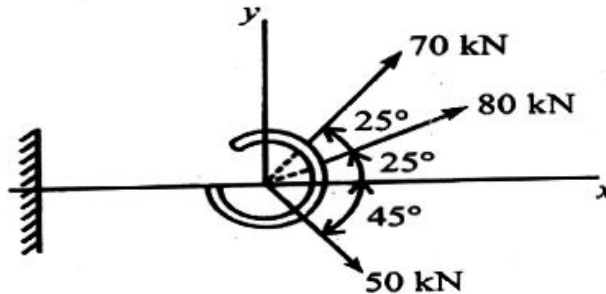
6X2=12M

1. a) Define (i) Line of action of force (ii) Resultant of force. 2M CO1 BTL1
- b) Mention the characteristics of friction. 2M CO2 BTL2
- c) State and explain pappu's theorem. 2M CO3 BTL3
- d) Define different Newton's law of motion. 2M CO4 BTL1
- e) Define the terms (i) Impulse momentum (ii) Coefficient of restitution. 2M CO5 BTL1
- f) State and explain varignon's theorem. 2M CO1 BTL2

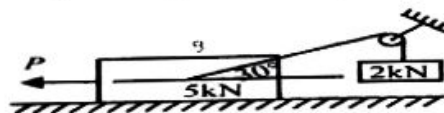
PART B

4x12=48M

2. Determine the resultant of the force acting on a hook as shown in figure 12M CO1 BTL4

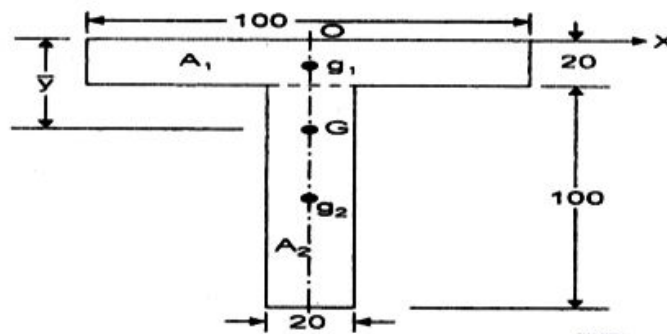


3. A block weighing 5kN is attached to a chord, which passes over a frictionless pulley, and supports a weight of 2kN. The coefficient of friction between the block and the floor is 0.35. Determine the value of force P if (i) The motion is impending to the right (ii) The motion is impending to the left. 12M CO2 BTL3



4. Locate the centroid of a given T – Section.

12M CO3 BTL3



5. a) Explain the terms (i) Kinetics (ii) Kinematics.

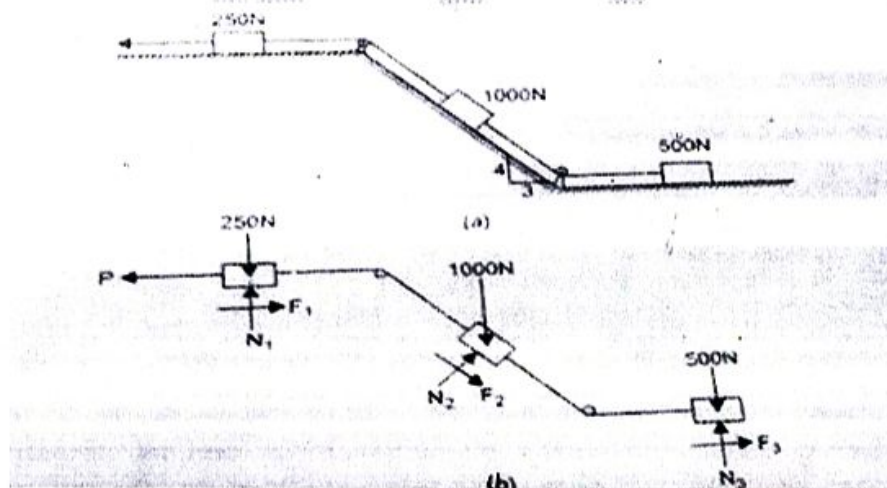
2M CO4 BTL2

b) A vertical lift of total mass 500Kg acquires an upward velocity of 2m/sec over a distance of 3m of motion with constant acceleration, starting from rest. Calculate the tension in the cable supporting the lift. If the lift while stopping moves with a constant deceleration and comes to rest in 2sec, calculate the force transmitted by a man of mass 75kg on the floor of the lift during the interval.

10M CO4 BTL2

6. Determine the constant force P that will give the system of the bodies shown in figure below a velocity of 3 m/s after moving 4.5 m from rest. Coefficient of friction between block and plane is 0.3. pulleys are smooth

12M CO5 BTL2



7. a) State and prove parallelogram law of forces.

6M CO1 BTL2

b) Discuss the following terms with suitable examples and diagrams.

6M

(i) Static friction (ii) Angle of repose (iii) Limiting force of friction.



Hall Ticket No. -

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Stanley College of Engineering & Technology for Women (A)

B.E (ECE/EEE) II Semester (Main) Examinations-September-2023

Engineering Physics

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
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Part-A

6X2=12M

- a. In a Newton's rings experiment, the diameter of the 15th ring was found to be 0.59 cm and that of the 5th ring is 0.336 cm. If the radius of curvature of the lens is 100 cm, find the wavelength of the light. [2M CO1 BTL4]
- b. List out the characteristics of Lasers. [2M CO2 BTL1]
- c. Define 'space lattice' and 'unit cell'. [2M CO3 BTL1]
- d. Give the physical significance of wave function ψ . [2M CO4 BTL2]
- e. State Ampere's law. [2M CO5 BTL1]
- f. Distinguish between Fresnel and Fraunhofer diffraction. [2M CO1 BTL3]

Part-B

4X12=48M

- a) What is interference of light? Explain in detail Young's double slit experiment and obtain the maxima and minima conditions. [6M CO1 BTL2]
- b) What is diffraction grating? Explain the experiment to determine the wavelength of a monochromatic light using diffraction grating. [6M CO1 BTL2]
- a) Describe the construction and working of He-Ne laser with the help of neat diagram by giving energy level transitions. [7M CO2 BTL3]
- b) Describe fibre drawing process by Double crucible method [5M CO2 BTL2]
- a) Define Miller indices. Deduce the expression for inter planer spacing for a cubic crystal system. [6M CO3 BTL4]
- b) Derive an expression for concentration of Schottky defects in ionic crystal [6M CO3 BTL3]
- a) Obtain an expression for energy of a particle in a 1-D box with the help of Schrodinger equation. [8M CO4 BTL4]
- b) State and prove Gauss law in electrostatics. [4M CO4 BTL2]
- a) Define Magnetic induction. State and explain Biot Savart's law. [6M CO5 BTL3]
- b) Derive an equation for Electromagnetic plane wave in a free space. [6M CO5 BTL4]
- a) Describe the powder diffraction method to calculate the lattice constant of a given crystal. [7M CO3 BTL3]
- b) What is optical fibre? Discuss types of optical fibres based on refractive index profile. [5M CO2 BTL2]

Stanley College of Engineering & Technology for Women (A)

B.E (Common to all) II Semester (Main) Examinations-September-2023

Mathematics-II

Time: 03Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any four out of remaining six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question Paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Solve $(y^2 - 2xy) dx + (2xy - x^2) dy = 0$ [2M CO1BTL2]
 b. Solve $(D^3 - 3D + 2) y = 0$ [2M CO2 BTL3]
 c. Evaluate $\int_0^1 x^5(1-x)^3 dx$ [2M CO3 BTL4]
 d. Define Rank of Matrix with example. [2M CO4BTL1]
 e. Find $L \{ t^2 \sin 2t \}$ [2M CO5 BTL3]
 f. Find the value of k such that the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2. [2M CO4 BTL2]

Part-B

4X12=48M

2. a) Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$ [6M CO1 BTL3]
 b) Prove that the system of parabolas $y^2 = 4a(x+a)$ is self-orthogonal. [6M CO1 BTL4]
3. a) Solve $(D^2 + 3D + 2) y = 2 \cos(2x+3) + 2e^x + x^2$ [6M CO2 BTL4]
 b) Solve the differential equation $(D^2 + 4) y = \sec 2x$ by the method of variation of parameters [6M CO2 BTL3]
4. a) State and Prove relation between Beta and Gamma functions [8M CO3 BTL4]
 b) Evaluate $\int_0^\infty x^2 e^{-x^2} dx$ [4M CO3 BTL3]
5. a) Determine whether the following equations have a non-trivial solution, if so solve them: $4x + 2y + z + 3w = 0$, $6x + 3y + 4z + 7w = 0$, $2x + y + w = 0$ [6M CO4 BTL3]
 b) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$ [6M CO4 BTL4]
6. a) Evaluate $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$ [4M CO5 BTL3]
 b) Evaluate $L^{-1} \left\{ \frac{s^2}{(s^2+4)(s^2+25)} \right\}$ using convolution theorem. [8M CO5 BTL4]
7. a) Solve $P = \sin(y - xp)$. Also find its Singular solution. [4M CO2 BTL2]
 b) Find the nature of the quadratic form $2x^2 + 2y^2 + 2z^2 + 2yz$ [8M CO4 BTL3]

Hall Ticket No. -

Code: R123931

Stanley College of Engineering & Technology for Women (A)

B.E (ECE/EEE/IT) II Semester (Main) Examinations-September-2023

Essence of Indian Traditional Knowledge

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. "Culture is diverse" Explain. [2M CO1BTL1]
b. Who is the author of "Pruthviraj Raso" and what is its specialty? [2M CO2 BTL1]
c. Write about Ramakrishna mission. [2M CO3 BTL2]
d. Explain how Kalighat painting got the name. [2M CO4 BTL2]
e. Name any four women scholars of Vedic era. [2M CO5 BTL1]
f. What is the meaning of the Sanskrit word "Itihasa". [2M CO2 BTL1]

Part-B

4X12=48M

2. a) What is unique about Indian Culture? [6M CO1 BTL2]
b) What is the impact of culture on Human life? [6M CO1 BTL2]
3. a) Explain the five characteristics of Puranas. [6M CO2 BTL1]
b) Write about the Telugu language and literature. [6M CO2 BTL2]
4. a) What were the religious beliefs of Pre-Vedic and Vedic people? [6M CO3 BTL2]
b) Assess the contribution of Arya-Samaj. [6M CO3 BTL2]
5. a) Write a short note on Medieval Indian Painting. [6M CO4 BTL1]
b) Write a short note in Indian classical music. [6M CO4 BTL1]
6. a) Describe the salient features of the NEP 2019. [6M CO5 BTL1]
b) Discuss the development of Science and Technology in Modern India. [6M CO5 BTL2]
7. a) What is unique about the Indian Architecture? [6M CO4 BTL2]
b) Write a short note on Ancient Indian Culture. [6M CO1 BTL1]

Hall Ticket No. -

Code: R123932

Stanley College of Engineering & Technology for Women (A)

B.E (ECE/EEE/IT) II Semester (Main) Examinations-September-2023

Indian Constitution

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Write about Preamble. [2M CO1 BTL2]
- b. Explain about qualifications of President. [2M CO2 BTL3]
- c. Explain about Right to Equality of Indian Constitution. [2M CO3 BTL1]
- d. Write a brief note on Central-State relations. [2M CO4 BTL3]
- e. What is the Composition of NCW? [2M CO5 BTL2]
- f. Why do we need Voting Right in Democracy? [2M CO1 BTL3]

Part-B

4X12=48M

2. a) What are the Features of Govt of India Act 1909? [6M CO1 BTL1]
- b) What was the composition of Constituent Assembly? [6M CO1 BTL2]
3. a) Explain about President of India. [8M CO2 BTL3]
- b) Explain about Governor of India. [4M CO2 BTL2]
4. a) Explain about Fundamental Rights. [8M CO3 BTL2]
- b) Explain the difference between Fundamental Rights and Duties. [4M CO3 BTL3]
5. a) Explain about Relations between Union and State. [5M CO4 BTL2]
- b) Explain the Composition and Objectives of NITI Aayog. [7M CO4 BTL3]
6. a) Explain the Powers and Functions of Election Commission of India. [8M CO5 BTL3]
- b) Explain about Powers of National Commission for Women. [4M CO5 BTL1]
7. a) Write an essay on the Salient Features of Indian Constitution. [8M CO1 BTL3]
- b) Explain about 74th Constitutional Amendment Act. [4M CO2 BTL2]

Stanley College of Engineering & Technology for Women (A)

B.E(ECE,EEE,IT) II Semester (Main) Examinations-September-2023

English

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
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Part -A

6X2=12M

2M L1 CO1

1. Do as directed.

- a. i) I am _____ university student. (an/a/the/no article)
 ii) The teacher congratulated Ravi _____ his success. (Write a suitable preposition.)
 iii) _____ in the workplace is harmful to its growth. [Root word : Loyal] (Write the apt word by using suffix-prefix)
 iv) Leela is the brightest student among all. (Change from superlative to positive degree of comparison)
- b. i) The boy is not meritorious. _____ he is very hard working. (Use suitable transition word)
 ii) _____ he is very famous, Azim Premji's modesty is exemplary. (Use an apt connective.)
 iii) Identify the root words in the given blended expression **Infotainment**.
 iv) Prof. Hary _____ from Australia yesterday. (Choose the suitable verb form.) has returned/ returned/ has been returning/ return back
- c. **Direct speech ⇌ Indirect speech**
 i) The teacher said to the student, "Do not waste time before your exam."
 ii) I told my supervisor that I would be finishing the task the next day.
- d. **Active voice ⇌ Passive voice**
 i) Beena is humming a beautiful tune.
 ii) The brinjal will be finely sliced by the chef.
- e. **Correct the given sentences.**
 i) Neither she nor her friends is interested in sports.
 ii) Each of the girls have precipitated in the game.
 iii) Salesman/Saleswoman (Find an inclusive term.)
 iv) The little boy is **handicapped**. (Write a suitable euphemism.)
- f. i) Make **prudent** use of your time. (Choose a suitable synonym.)
 I. Judicious II. Mandatory III. Trivial IV. Imbecile
 ii) Mr. Alex is a **pretentious** art critic. (Choose a suitable antonym.)
 I. Flatulant II. Hollow III. Genuine IV. Overambitious
 iii) The red rose tucked in your open hair _____ your saree.
 (Compliments/Complements)
 iv) Government by the people is known as _____. (Write the one-word substitute.)

2M L3 CO3

2M L3 CO4

2M L3 CO5

2M L1 CO3

Part-B

4X12=48M

2. i) William Hazlitt feels that being in a school /hostel will teach his son about how to get along with others and prepare him for the ups and downs of life. Do you agree to the opinion? Analyse with textual references. 6x2M L4 CO1
L2
- ii) Explain any four essentials to draft an effective paragraph.
3. i) "If you can meet with Triumph and Disaster
And treat those two impostors just the same;"
Who is the author of the poem? What is the meaning of the word 'imposter'? Analyse the significance behind labelling 'triumph' and 'disaster', 'two imposters.' 1+1+4 M L1, L4 CO2
- ii) Outline the differences between expository and descriptive writing style? 3+3 M
Read the following paragraph and determine whether it is an expository or a descriptive paragraph. Give reasons to your answer. L2, L3
The terrestrial planets are made of rock and metal and are closest to the sun. These include the midsize planets Mercury, Venus, Earth, and Mars. They rotate slowly and don't have many moons. Farther from the sun are the planets called gas giants, Jupiter, Saturn, Uranus, and Neptune. They are called gas giants because they are formed from gases such as hydrogen and helium.
4. i) Explain a few strategies to craft a persuasive paragraph. 6 M L2 CO3
ii) What does the term "Dream" refer to in Martin Luther King's speech? Examine if this famous speech holds its relevance in today's time too. Justify your answer with relevant examples. 3 + 3 M L4
5. i) "I took the one less travelled by, // And that has made all the difference." Who says this? Extend your views on the significance of these two lines. 6X2M L1, L2 CO4
ii) Compose an argumentative essay on "Has the internet made our society better?" L6
6. i) Discuss any four characteristics of a successful blog. 4+ 8 M L2 CO5
ii) Justify how writing on Instagram or Twitter is different from academic writing. Has writing on social media platforms impacted academic writing? Illustrate with examples. L5
7. i) As an effective communicator, why should you learn the uses of Inclusive language and Euphemism? Cite 3 examples for each. 6x2M L2 CO5
ii) Develop a coherent paragraph by using the given hints: L6 CO1
Giving pledge – Philanthropic cause – launched by Bill Gates– many are unprivileged – need support – the wealthiest people can share – make the world a better place – social responsibility